AEGIS Verification Platform

Month 2 Milestone: Verification Modes UI Polish & Architecture Guide

Target Audience: Evaluators, Stakeholders & Devs	**Branch:** feat/polish-verification-modes-ui	**Status:** Validated & Ready (PR #66)

1. Executive Summary (The Simple Big Picture)

AEGIS is an agentic cybersecurity platform built to detect **AI-generated deepfakes, synthetic speech, and camera stream hijacking**. The system provides two verification modes: **Mode 1 (Offline File Analysis)** for forensic investigation of recorded media, and **Mode 2 (Live Surveillance)** for continuous monitoring of security cameras and live video feeds.

Why this task was needed: User testing during Week 3 revealed three key friction points: (1) Mode 1 and Mode 2 felt like disjointed, separate applications; (2) Alert severity was vague, leaving analysts unsure whether an alert was an emergency or a benign glitch; and (3) Demonstrating the platform required manual file uploads and complex setups.

What we delivered: We executed a focused polish pass creating a unified visual language, standardized 3-tier SIEM severity badges, integrated the Month 2 multi-agent Judge Agent debate timeline, added live attack injection, and provided 4 one-click demo presets.

2. Standardized Severity Language: SIEM Alert Tiers

To ensure security operations centers (SOC) can triage threats instantly, we introduced standardized 3-tier severity badges:

SIEM Tier	Visual Indicator	Plain-English Meaning	Action & System Behavior
TIER 1 LOW	**GREEN SHIELD**	Media is verified authentic. All AI detectors and biometrics agree with high confidence.	Standard logging. Certified court-admissible authentic media fixture.
TIER 2 ELEVATED	**AMBER TRIANGLE**	Detectors disagree, media has high compression, or biometric pulse quality is low.	Flagged for secondary manual inspection by a human forensic analyst.
TIER 3 CRITICAL	**ROSE SIREN**	Confirmed synthetic manipulation, acoustic voice cloning, or active camera spoofing.	Immediate SIEM alert dispatch, audio-visual strobe siren, and operator review required.

3. Mode 1: Offline File Verification Flow

Analogy: Like sending evidence to a digital forensics lab. An investigator submits an audio or video file (or selects a preset). The system executes a deep multimodal analysis and synthesizes the findings into a verifiable forensic dossier.

Key Features Implemented & Polished:

- **Deterministic Judge Agent:** An impartial algorithmic judge synthesizes detector claims, evaluates modality capabilities, and produces an authoritative court-ready verdict.
- **3-Round Debate Visualizer:** Users see transparent reasoning across *Round 1 (Initial Modal Claims)*, *Round 2 (Cross-Modal Cross-Examination)*, and *Round 3 (Judge Synthesis)*.
- **Safety Guard Execution:** Hardcoded safety rules protect against errors. **Guard 1 (Polarity Inversion Anti-Default)** prevents false-authentic defaults when cloned audio accompanies genuine video.
- **Tamper-Evident SHA-256 Hash:** Cryptographically locks the media and transcript with a 1-click clipboard copy button.

4. Mode 2: Live Surveillance Monitoring Flow

Analogy: Like an intelligent security guard watching airport CCTV. Mode 2 connects to live camera feeds, executing rapid sliding-window analysis every 200 milliseconds to catch deepfake face replacements or video stream hijacking in real time.

Key Features Implemented & Polished:

- **Detailed Evidence Modal:** Clicking 'Inspect Evidence' on the alert banner pops open a complete breakdown of sensor telemetry (temporal consistency, PRNU camera sensor fingerprint, and rPPG blood volume pulse).
- **Interactive Attack Injection ([INJECT ATTACK]):** Evaluators can simulate active sensor degradation and deepfake injection on demand, demonstrating immediate siren triggers and Tier 3 alert generation.
- **Operator Acknowledgment & Audit Trail:** Security personnel can click 'Acknowledge' to silence alarms and log an immutable record of operator incident handling.
- **Harmonized Navigation Switcher:** Both the standalone workstation and Next.js shell now clearly indicate 'Offline File [Single-Pass]' versus 'Live Surveillance [Sliding-Window Stream]'.

5. Curated 1-Click Milestone Demo Presets

To make testing seamless for evaluators without needing sample uploads, we built four 1-click evaluation fixtures:

Preset Scenario	Modality	What it Demonstrates (Plain English)	Expected Tier
Clean Baseline *clean_interview_broadcast.mp4*	Video + Audio	Unedited natural interview. All 4 detectors (visual, rPPG pulse, SyncNet, AASIST) agree authentic.	**TIER 1** LOW
Voice Clone Scam *synthetic_voice_clone_scam.wav*	Audio Only	AASIST flags vocoder acoustic synthesis. Visual models are safely bypassed via capability manifest.	**TIER 3** CRITICAL
Silent Surveillance *surveillance_silent_clip.mp4*	Video Only	Zero-audio CCTV footage. Pre-dispatch filter safely skips audio models and verifies camera textures.	**TIER 1** LOW
Cross-Modal Conflict *dubbed_speech_conflict.mp4*	Video + Audio (Dubbed)	Genuine face but cloned audio. Guard 1 triggers, preventing false-authentic default and escalating to Tier 3.	**TIER 3** CRITICAL

6. GitHub CI / Pull Request #66 Issue & Resolution

When submitting PR #66 to GitHub, the automated check **'lint-and-build'** failed at the **Run Impeccable UI Linter** step. Here is what happened and how we resolved it:

- **The Linter Finding:** The design linter flagged animate-bounce on line 243 of live/page.tsx. Professional UI standards discourage bouncy, springy animations on technical security buttons because they look dated and distracting.
- **The Engine Mismatch:** The linter package required Node.js >=22.18.0, whereas GitHub Actions was pinned to Node.js 20.
- **The Resolution:** We replaced the bounce animation with a smooth, professional animate-pulse effect and bumped GitHub Actions to Node.js 22. The linter passed locally with **0 anti-patterns**.

7. Comprehensive Quality & Verification Results

Every single layer of the application was tested and verified prior to branch merge:

Verification Test Suite	Scope Covered	Result
Next.js Production Build npm run build	All 19 static and dynamic routes (/analyze, /live, /reports/[id], /api/jobs)	**PASSED (19/19)** Zero TS/JSX errors
Debate & Judge Agent pytest debate/	Multi-agent arbitration, schemas, disagreement detection, and Month 1-2 cases	**PASSED (86/86)** 4.03s execution
Orchestrator Pipeline pytest orchestrator/	Judge integration, wiring, capability manifest filtering, and telemetry client	**PASSED (68/68)** 100% clean
Security & Alert Routing pytest security/tests/	SIEM alert dispatch, fusion harness, and stream validator	**PASSED (13/13)** All tests green
Live Server Probe curl http://localhost:3000/...	End-to-end HTTP response codes on all primary endpoints and API routes	**PASSED (200 OK)** Full JSON metadata

8. How Anyone Can Experience the Demo in 2 Minutes

To evaluate the Month 2 milestone live, follow these two simple steps:

- **Step 1: Test Offline Deepfake Adjudication (Mode 1)**
  Open http://localhost:3000/analyze. Click **'Report'** on the **Cross-Modal Conflict** preset. Observe the **SIEM TIER 3 — CRITICAL** badge, the **Guard 1** pill, the 3-round debate timeline, and click **Copy** on the tamper-evident SHA-256 hash.
- **Step 2: Test Live Camera Surveillance (Mode 2)**
  Switch to **Live Surveillance** (or visit http://localhost:3000/live). Click **[INJECT ATTACK]**. Watch the live risk score surge, the red siren strobe trigger, and click **'Inspect Evidence'** to view the fused sensor modal before clicking **'Acknowledge'**.

9. Conclusion

The verification-mode interfaces are now unified, visually polished, and battle-tested. With deterministic Judge Agent reasoning, tamper-evident cryptographic audits, and reliable one-click demonstration fixtures, AEGIS is fully ready for the Month 2 milestone demo.